Underlying idea

Causal interpretation of hypotheses requires one of the following strategies:

Variant 1 (longitudinal)

Removed from this PAP

Variant 2 (quasi-experimental)

Quasi-experimental design: **gender of the first child may be considered a random event**, therefore the differences in outcome between parents having a daughter as their first child vs. those having a son may be considered a consequence of this random event.

Note that whereas gender of the first child may be considered random, chances of having any daughter may not. Specifically, the probability of ever having a daughter may depend on the outcome of interest (e.g., gender-traditional parents may be more likely to have an (additional) child, thus they may be more likely to have a daughter). For this reason, in the quasi-experimental setup the total number of daughters cannot be used to estimate the causal effect of a child's gender on the outcome.

We include parents **independently of the number of children** and we do not limit the analysis to parents having only one child because (assuming that total fertility may be determined by the outcome, e.g., gender attitudes) that would imply conditioning the inclusion in the analytical sample on the outcome of interest.

Moreover, we **only focus on the gender of the first-born child**, and do not consider the gender of subsequent children. If the gender of the first child is a random event, then there should not be systematic differences between parents having a daughter vs. son as first-born child. Thus, also subsequent fertility should not differ between these groups. If fertility patterns differ between these groups, then they should be considered a consequence of the gender of the first child. This speaks against controlling for subsequent fertility, which may be considered a factor mediating part of the effect. Thus, one could speak of a **direct effect of first-born child's gender** on the outcome, and the **indirect effect, mediated by subsequent fertility**.

Pre-analysis plan, hypothesis 1

H1. *Having daughters leads to stronger support for gender equality (as measured by attitudes).*
That is, denoting X as a measure of support for gender equality,
(X for respondents with daughters) > (X for respondents without daughters).

H1 Variables

H1 Dependent variable

SOEP original variables

We identified three variables measuring attitudes capturing support for gender equality which were measured more than once (see the table below).

Question	Variables	Year	Valid N*	Answering scale
("Suffer6") A child aged under 6 will suffer from having a working mother. (Ein Kind unter 6 Jahren wird darunter leiden, wenn seine Mutter arbeitet.)	plh0298_v1	2012	7131	**2012:** [1] Agree fully [2] Rather agree [3] Rather disagree [4] Fully disagree **2018:** [1] Fully disagree [2] [3] [4] [5] [6] [7] Agree fully
	plh0298_v2	2018	25527	
("Suffer3") A child aged under 3 will suffer from having a working mother. (Ein Kind unter 3 Jahren wird darunter leiden wenn seine Mutter arbeitet.)	plh0302_v1	2012	7117	
	plh0302_v2	2018	25570	
("SameShare") Best if man and woman work the same amount so they can share the responsibility (Am besten ist es, wenn der Mann und die Frau beide gleich viel erwerbstätig sind und sich beide in gleichem Maße um Haushalt und Familie kümmern)	plh0308_v1	2012	7072	
	lh0308_v2	2018	25570	

* According to paneldata.org. The 2012 data come from the *Familie in Deutschland* study and seems not to include any childless people (maybe with the exception of low-income households; Geue et al., 2020).

Among the three variables, the variable "SameShare" closely covers the concept of support for gender equality, whereas "Suffer3" and "Suffer6" variables measure the support for traditional mothers' role, which is a related but distinct concept from support for gender equality.

To measure the support for gender equality, we might rely exclusively on "SameShare." However, we would then be forced to rely on a limited dependent variable measured on a 4-point scale (in 2012), which is cumbersome with models for continuous outcomes.

Therefore, we prefer to rely on an index incorporating information from all three variables. By doing this we assume that, when answering all three questions, the respondents had in mind the idea of sharing paid work and housework within a couple.

Attitudes supporting gender equality – index construction

We aim to find a transformation which allows us to compare the values of the 2012 and 2018 variables, preserving the variance. (Alternative specifications, which do not preserve the variance, include (a) standardizing the variables, which would impose an equal variance, (b) stretching the scales to the same range, e.g. 1-7 or 1-10, which would stretch the variance of values in 2012.)

We plan to construct the index of support for gender attitudes ("GEIndex") by following these steps:

1. **Transform the values** (1-4 or 1-7) into **percentile scores**, which correspond to the sum of:
    a. The percentage of the sample who support gender equality LESS than the respondent
    b. Half of the size (expressed as a percentage) of the category to which the respondent belongs

    E.g. for "SameShare" variable in 2012 this transformation could look like that:

Answers	Hypothetical relative frequencies	Hypothetical percentile scores
[1] Agree fully (highest support)	20%	80% + ½ * 20% = 90%
[2] Rather agree	30%	50% + ½ * 30% = 65%
[3] Rather disagree	40%	10% + ½ * 40% = 45%
[4] Fully disagree (lowest support)	10%	½ * 10% = 5%

2. Calculate a **weighted sum** of the percentile scores to construct the index (allowing max one missing value for each respondent). Considering that the variables "Suffer3" and "Suffer6" have similar meaning, whereas "SameShare" has a different meaning, most closely approaching support for gender equality, we differentiate the weights:
    a. Weight of 0.25 for "Suffer3" and "Suffer6"
    b. Weight of 0.5 for "SameShare."

H1 Main predictor

To assess how many children and of which gender a respondent had in 2012 and 2018, we will use the fertility histories as recorded in BIOBIRTH data.

Main predictor: first child is a daughter.

H1 Control variables

Validity of our approach depends on the independence of sex ratio at birth from the outcome of interest (in H1, support for gender equality as captured by attitudes). The Trivers-Willard hypothesis postulates that sex-ratio at birth may differ with parental characteristics, so that

more girls are born in conditions that would favor the reproductive success of women. Consistently with this idea, research linked higher chances of girl's birth with physical attractiveness of parents (Kanazawa, 2011) and working in so-called "empathizing" (vs. "systemizing") occupations (Kanazawa & Vandermassen, 2005). Other research focused on factors that could more closely be linked with support for gender equality. In this literature, the hypothesis that older mothers have a higher chance of birthing a daughter received mixed support (supported by Matsuo et al., 2009 but refuted by Rueness et al., 2012). Similarly, the hypothesized link between sex ratio at birth and socioeconomic status of mothers has not been supported by empirical studies (Houdek et al., 2020; Kolk & Schnettler, 2016; Sarkadi et al., 2024). Considering this evidence, we conclude that the chance of birthing a girl does not differ with the level of support for gender equality.

Considering this, we opt for a **minimum list of controls** in Variant 2.

Concept and values	Justification of the control variable	Variable
Year of birth of the 1st child (linear and quadratic term to allow for non-linear effects)	(a) If a mother's age at first birth has been increasing over time, then this should have increased the probability of birthing a daughter. (b) Support for gender equality has plausibly increased over time.	From BIOBIRTH file grand mean centered

We do not control for mother's age at first birth, because, although more advanced age may increase the chance of birthing a daughter, it probably does not affect support for gender attitudes. (We assume that, if such a correlation exists, it probably reflects the effect of attitudes on the age at first birth, and not the other way round.)

H1 Other variables used in sample selection and analysis

Concept and values	Where used	Variable
ID of respondents	To define a longitudinal data for FE analysis (xtset)	pid
Survey year	To define a longitudinal data for FE analysis (xtset) To create the year dummy	syear
Age of respondent	Sample selection	ppathl/ gebjahr Age = syear – gebjahr
Sex of respondent	Sample selection	ppathl/sex
Heterosexual	Sample selection	ppathl/sexor

H1 Inclusion/exclusion criteria

(1) Respondents observed in **2012 OR 2018** with valid answers to questions on gender attitudes.
(2) Who had **at least 1 child** at the time of observation.
(3) Who, at the time of observation, where **18+** years old.
(4) Who are **heterosexual and gender binary**, because the notion of gender equality refers primarily to the gender division of (paid and unpaid) work in heterosexual couples.

H1 Exact analysis

The "natural-experiment" setup will use a model with individual random effects.

$$\text{GEIndex}_{it} = \alpha + u_i + \beta_1 FirstDaughter_{it} + \beta_4 BirthYear_{it} + \beta_5 BirthYearSq_{it} + e_{it},$$

$$u_i \sim N(0, \sigma^2)$$

```
xtreg ge_index daughterfirst daughterYB daughterYBSq, re
```

H1 Interpretation of results

The coefficient of variables $FirstDaughter_{it}$ informs how many percentiles up/down in the support for gender equality (expressed by attitudes) does having a daughter as the first child propel the respondent, compared to respondents who had a son as their first child.

We will conclude that H1 has been supported if:

- the coefficient of $FirstDaughter_{it}$ $\beta_1 > 0$,
- $p < 0.05$ (because of large sample),
- In case of statistically significant results, we will substantively interpret the size of effects.

Pre-analysis plan, hypothesis 2

H2. *Having daughters leads to more left-leaning political preferences.*
That is, denoting Y as a measure of left-leaning political preferences,
(Y for respondents with daughters) > (Y for respondents without daughters).

H2 Variables

H2 Dependent variable

The SOEP variable **plh0004** (pl file) collected answers to a question *"In politics people often talk about "left" and "right" when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours?"* and recorded them on a scale from 0 – „far left" („ganz links") to 10 – „far right" („ganz rechts").

The variable was recorded for waves 2005, 2009, 2014, and 2019.

As the hypothesis refers to left leaning attitudes, we will:

- reverse the scale so that higher values correspond to more left-leaning attitudes, and
- stretch the scale (for easier interpretation as percentage points) so that 100 means "far left" and 0 means "far right."

H2 Main predictor

See Hypothesis 1.

H2 Control variables

As in Hypothesis 1: control for year of birth of the first child.

H2 Other variables used in sample selection and analysis

See Hypothesis 1.

H2 Inclusion/exclusion criteria

(1) Respondents observed in **at least one wave among 2005, 2009, 2014, and 2019** with valid answers.
(2) Who had **at least 1 child** at the time of observation.
(3) Who, at the time of the observation, were **18+** years old.
(4) Who are **heterosexual and gender binary** to keep the group relatively homogenous.

H2 Exact analysis

The "natural-experiment" setup will use a model with individual random effects.

$$\text{LeftLeaning}_{it} = \alpha + u_i + \beta_1 FirstDaughter_{it} + \beta_4 BirthYear_{it} + \beta_5 BirthYearSq_{it} + e_{it},$$

$$u_i \sim N(0, \sigma^2)$$

```
xtreg Left daughterfirst daughterYB daughterYBSq, re
```

H2 Interpretation of results

As in Hypothesis 1.

Pre-analysis plan, hypothesis 3

H3. *Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).*
That is, denoting Z as a measure of gender equality in the household,
(Z for respondents with daughters) > (Z for respondents without daughters).

H3 Variables

H3 Dependent variable

As a behavioral measure of gender equality, we consider the **between-partners similarity in time allocation** within heterosexual couples.

Original variables

SOEP records a set of variables for different types of activities; for the typical weekday, Saturday and Sunday separately.

	Weekday	Saturday	Sunday
PAID WORK / EDUCATION AS AN INVESTMENT IN PAID WORK			
Job, apprenticeship, second job / Job, Training Beruf, Lehre, Nebenerw	pl/pli0038_h 1984-2022	pl/pli0003_h 1990, 2012 + Odd waves 1993-21	pli0007_h 1984-1990, 1992, 2012 + Odd waves 1993-21
Education or further training (also school / university) Aus- u. Weiterb., Lernen Std	pli0047_h 1984-2022	pli0024_h 1990, 2012 + Odd waves 1993-21	pli0028_h 1984-1990, 1992, 1996, 2012 + Odd waves 1993-21
UNPAIND WORK			
Errands (shopping, trips to government agencies, etc.) Besorgungen /Besorgungen und Einkaufen	pl/pli0040 1990-2022	pl/pli0054 1990, 2012 + Odd waves 1993-21	pl/pli0011 2012 + Odd waves 2001-21
Housework (washing / cooking / cleaning) Hausarbeit Std	pl/pli0043_h 1984-2022	pli0012_h 1990, 2012 + Odd waves 1993-21	pli0016_h 1984-1990, 1992, 2012 +

			Odd waves 1993-21
Childcare Kinderbetreuung	pl/pli0044_h 1985-2022	pli0019_h 1990, 2012 + Odd waves 1993-21	pli0022_h 1985-1990, 1992, 2012 + Odd waves 1993-21
Care and support of persons in need of care Versorgung Pflegebeduerftiger	pli0046 2001-2022	pli0055 odd waves 2001-2021	pli0057 odd waves 2001-2021
Repairs on and around the home, car repairs, garden or lawn work Reparaturen etc.	pli0049_h 1984-2022	pli0031_h 1990, 2012 + Odd waves 1993-21	pli0034_v4 1992, Odd waves 1993-21

Our analysis will cover the odd years 2001-2019 (i.e. 2001, 2003, 2005,.. 2019, summing up to 20 years and 10 time points), which allows observing the **weekly time spent in paid work / unpaid work / other (residual category).** We choose to focus on **weekly** (and not weekday) time allocation, because, due to regulatory limitations on working time, fathers have little choice during the week, and they may use the weekend to catch up on time allocation.

Similarity in time allocation – index construction

We plan to construct the index of similarity in time allocation ("SimilarityIndex") by following these steps:

1. For each respondent, calculate the **weekly hours spent on paid work**, as:
   5 x weekday hours in paid work + Saturday hours in paid work + Sunday hours in paid work
2. Calculate the **weekly hours spent on unpaid work**;
3. Calculate the **residual hours**, corresponding to 24h x 7days – weekly hours in paid work, – weekly hours in unpaid work;
4. Calculate the index of dissimilarity for each couple (White, 1986):

$$D = \frac{1}{2}\sum_{i=1}^{I}\left|\frac{n_{im}}{N_m} - \frac{n_{if}}{N_f}\right|$$

Where:

- i indexes types of activity (and there are I types of activity overall),
- m and f refer to male and female partners respectively,
- n_{im} refers to the hours men spend in activity i,
- N_{im} refers to the total hours available to men (in our case it is constant at 24h x 7days = 168 hours).

The score of the index corresponds to **the share of time that would need to be reallocated** to achieve equal time allocation within a couple.

5. Finally, we **reverse the scale** to 1-D, so that it measures the extent of similarity, and not dissimilarity. The resulting value should be interpreted as the share of the time which the partners allocate in the same way.

H3 Main predictor

See Hypothesis 1.

H3 Control variables

As in Hypothesis 1: control for year of birth of the first child.

H3 Other variables used in sample selection and analysis

As in Hypothesis 1. We will additionally use information on the id of the partner

Concept and values	Where used	Variable
ID of respondent's partner	To match the partners	ppathl/parid

H3 Inclusion/exclusion criteria:

In contrast to other hypotheses, in H3 the unit of analysis is a couple (not a respondent).

Sample will be selected as in Hypothesis 2, but considering different waves of interest (odd waves between 2001 and 2019).

H3 Exact analysis

As Hypotheses 1 and 2

H3 Interpretation of results:

As in Hypothesis 1.

Pre-analysis plan, hypothesis 4

H4. *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).*
That is, denoting X as a measure of support for gender equality,

$$\begin{pmatrix} X \text{ for male respondents with daughters} \\ - X \text{ for male respondents without daughters} \end{pmatrix} > \begin{pmatrix} X \text{ for female respondents with daughters} \\ - X \text{ for female respondents without daughters} \end{pmatrix}.$$

H4 Variables

The hypothesis builds on Hypothesis 1, additionally testing for gender differences. The variables will follow closely on variables constructed to test Hypothesis 1.

H4 Inclusion/exclusion criteria

See Hypothesis 1.

H4 Exact analysis

Compared to Hypothesis 1, the test of Hypothesis 4 introduces an interaction term between male gender of the respondent and gender of the first child.

The "natural-experiment" setup will use a model with individual random effects.

$$\text{GEIndex}_{it} = \alpha + u_i + \beta_1 FirstDaughter_{it} \times Man_i + \beta_4 BirthYear_{it} + \beta_5 BirthYearSq_{it} + e_{it},$$

$$u_i \sim N(0, \sigma^2)$$

```
xtreg ge_index daughterfirst##Man daughterYB daughterYBSq, re
```

H4 Interpretation of results

The coefficient of variables $FirstDaughter_{it} \times Man_i$ informs how many percentiles up/down in the support for gender equality (expressed by attitudes) does having a daughter as the first child propel men more than women.

We will conclude that H1 has been supported if:

- the coefficient of $FirstDaughter_{it} \times Man_i$ is greater than 0,
- $p < 0.05$ (because of large sample),
- In case of statistically significant results, we will substantively interpret the size of effects.

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